

AUTONOMOUS MAZE SOLVER ROBOT



ECOLE MAROCAINE DES
SCIENCES DE L'INGENIEUR

Membre de 
HONORIS UNITED UNIVERSITIES



Presentation Outline



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Added Values

Autonomous Maze Solver Robot

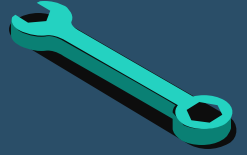
02





01

Objectives

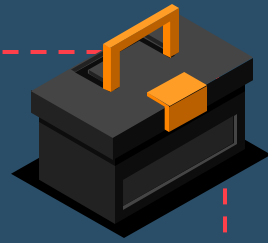


03

Autonomous Maze Solver Robot

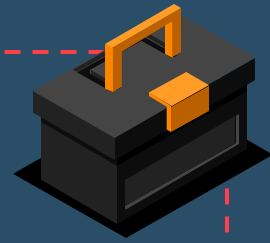
Starting Sequence

Upon startup, the robot displays "Welcome! I'm DAMI 3000!" with an audible and visual signal using the RGB LED module. It then enters a menu where the user can choose between several modes via the buttons in a 4×4 matrix. The 7-segment display then scrolls quickly, as a sign of awaiting selection.



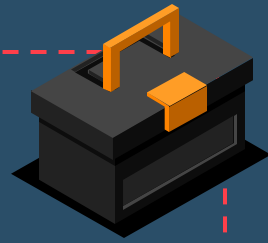
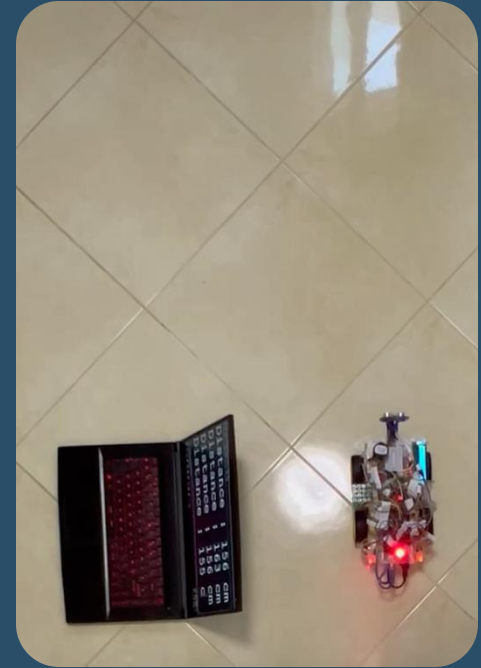
Maze Mode

In maze mode, the robot moves forward until it detects an obstacle at 15 cm, backs up for 0.5 seconds, then scans to the right and left with its servo motor. It always prioritizes turning right (Right-hand algorithm) if the space is clear (more than 20 cm). If the right side is blocked, it checks the left. If the left is also blocked, the robot performs a U-turn (180 degrees) to find a new path. This cycle repeats for navigation, and thus, to exit the maze.

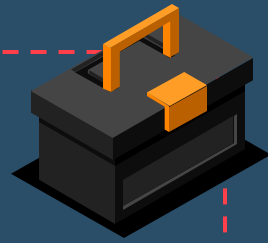


Bluetooth Mode

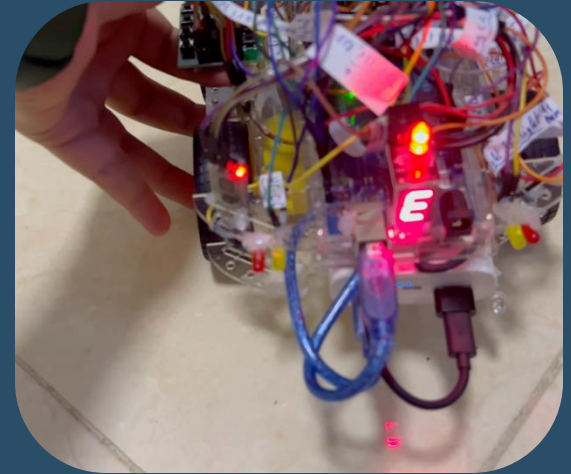
The robot is controlled remotely via Bluetooth. The keys Z, S, D, Q are used to steer it. The space bar is used to stop it, B to activate and deactivate the buzzer, and L, M, O, K to orient the Ultrasonic sensor. The LCD screen displays the sensor's status and distances. Red LED = inactive, Green LED = active. The emergency X button allows returning to the menu, stopping any robot movement, and centering the servo-motor. All of this was made possible using the HC-06 module connected to the board & CoolTerm on Windows.



Emergency Mode



Upon activation, it executes a predefined sequence of movements (forward, backward, left, right, stop) and servo motor operations (left, center, right positions), while integrating distance measurement via the ultrasonic sensor, beeping the buzzer, lighting up all pins of the 7-segment display, and the motion status LEDs (signal, stop, reverse). This is to determine if a component requires maintenance, or if a new prototype code has damaged a component, a cable has been disconnected, etc.



Each step is visually indicated on the LCD screen and by the RGB LED module.



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Components

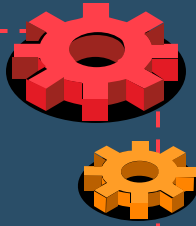


Autonomous Maze Solver Robot

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Components



Arduino Board – x1	HC-06 – x1
L298N – x1	Servo sg90 – x1
Chassis Pack – x4	HC-SR04 – x1
Switch – x1	LEDs – x4
Battery 18650 – x2	LCD+I2C 16*02 – x1
RGB Module – x1	Buzzer – x1
Joystick Module – x1	7 Seg CC – x1
18650 Charger TP4056 – x1	Matrix 4x4 – x1



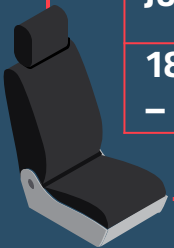
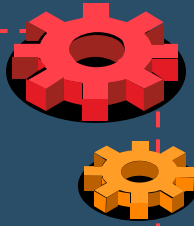
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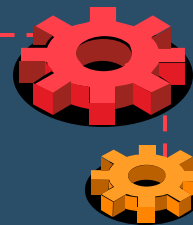
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RGB Module – x1	Buzzer – x1
Joystick Module – x1	7 Seg CC – x1
18650 Charger TP4056 – x1	Matrix 4x4 – x1

Stable system and cable management

The system manages its power through various energy sources

Resistors for every sensitive component for protection





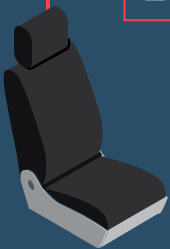
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RGB Module – x1	Buzzer – x1
Joystick Module – x1	7 Seg CC – x1
18650 Charger TP4056 – x1	Matrix 4x4 – x1

**Components
approximative price**

1000 Dhs

NB : We had to buy the needed tools, which added more cost to the budget ←

Autonomous Maze Solver Robot



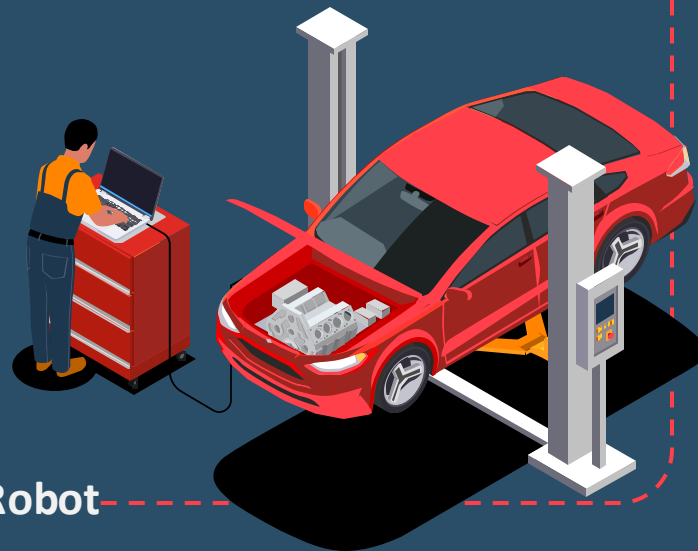


Made with
Fritzing



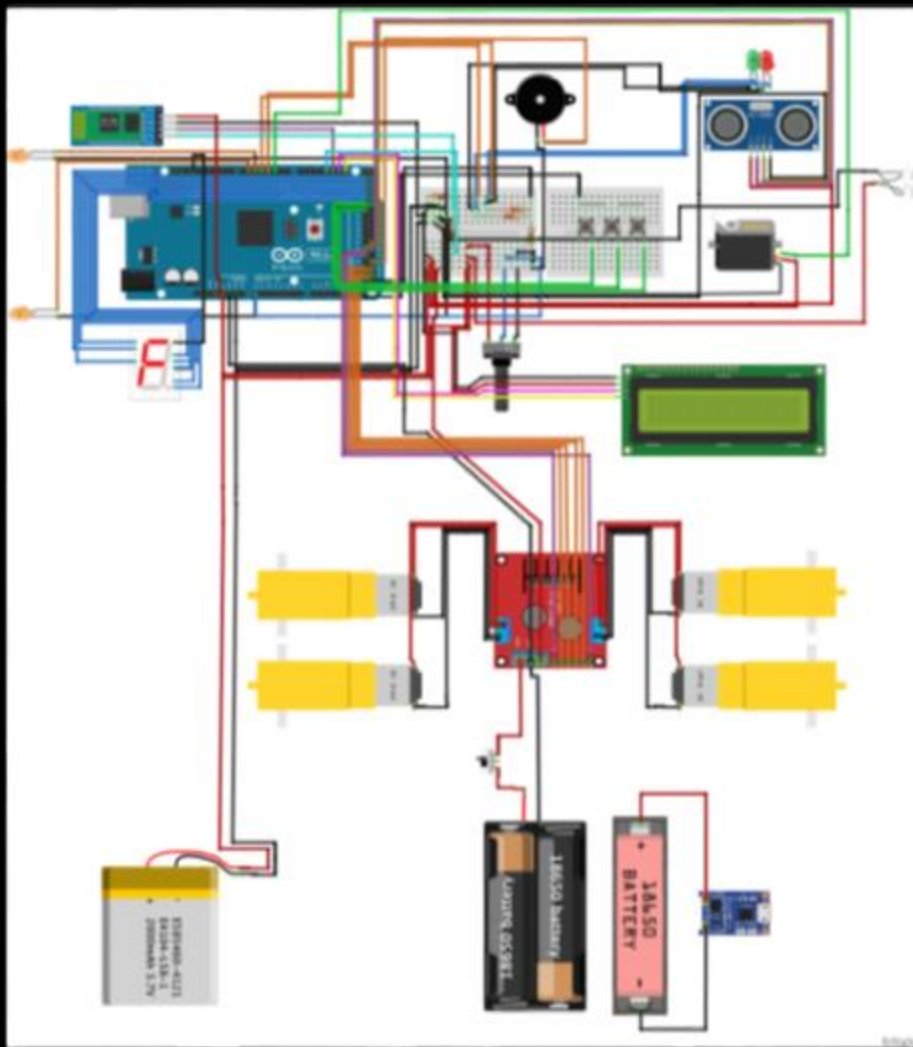
03

Wiring



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fritzing

each component was made
by the community of Fritzing

Fritzing is not made for
simulation, but rather for
the wiring design only

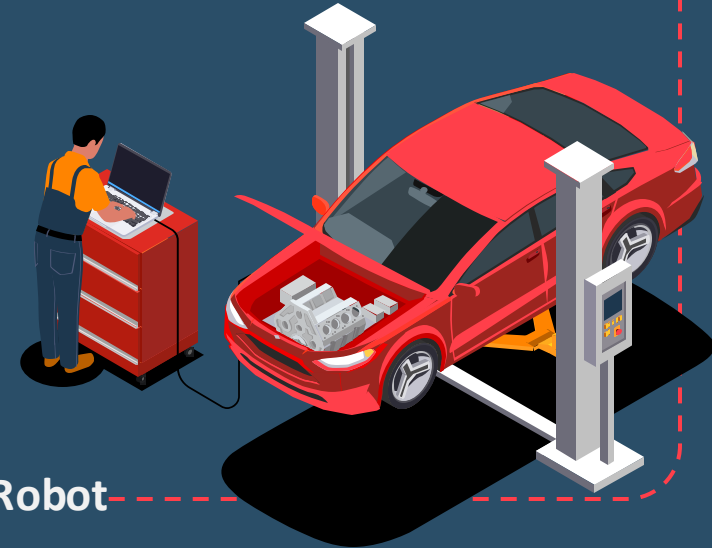
As you guys see, a LOT of
wires and resistors were
needed, which explains the
budget and the complexity



Made with
Arduino IDE



04 Coding

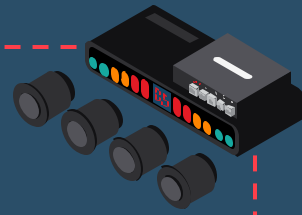


05

Added Values



Added Values



Interactivity

Buzzer, 7 seg, Servo movements, etc

Bluetooth Capabilities

Giving the user the total control over each component by itself

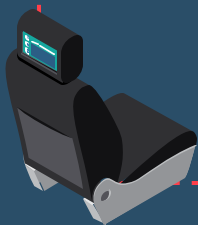
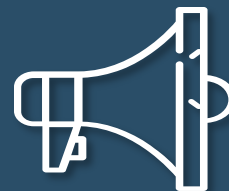
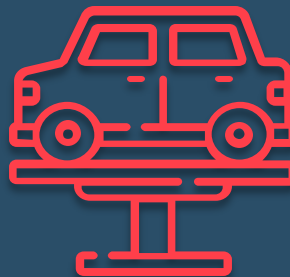
Energy management

Batteries, Powerbank, Rechargable, switch, etc

Dami 3000 is also programmed to make some movements with its head to make it seem more natural and playful



Testing Phase





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Thanks!

Do you have any questions ?



Presented & Created By :

Nizar El Idrysy
Zerrari Ziyad
Amine Habz
Bilal Mesbahi

